

TARUN R NAIK

Bengaluru, KA



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Email



LinkedIn



GitHub

Education

Nitte Meenakshi Institute of Technology (NMIT)

B.E. in Information Science & Engineering - CGPA: 6.65 / 10.0

Devika PU Science College

XII - Science (PCMB) - Percentage: 77.8%

Amrutha Varshini Vidyalyaya

X - Percentage: 81.9%

Bengaluru, KA

Sept 2023 – Aug 2027

Ranebennur, KA

June 2021 – March 2023

Ranibennuru, KA

TECHNICAL SKILLS

Languages: C, Python, Java

Backend & Data: MySQL, Pandas, NumPy, Scikit-learn

Machine Learning: Supervised & Unsupervised Learning, Time Series Forecasting

Web Technologies: HTML, CSS

Platforms: Git, GitHub, VS Code

INTERNSHIP

Data Science Intern

Princeton Smart Engineers

Internship Drive

Bangalore, KA | July 2025 – Sept 2025

- Developed a Machine Learning-based Vande Bharat Express occupancy forecasting system using Random Forest, KNN, SVM, and Decision Tree algorithms.
- Performed data preprocessing, exploratory data analysis (EDA), feature engineering, normalization, and categorical data transformation for railway occupancy prediction.
- Trained and evaluated multiple ML models using Accuracy, Precision, Recall, and F1-score, achieving 83.33% Accuracy, 69.44% Precision, 83.33% Recall, and 75.76% F1-score.
- Analyzed route-wise, temporal, and occupancy patterns using visualization and correlation analysis to support demand forecasting, capacity planning, resource allocation, and data-driven railway operations.

PROJECTS

Traffic Flow Prediction Using Graph Neural Networks (GNN)|

Python, PyTorch, Streamlit, FastAPI, Leaflet, TomTom API, Graph Neural Networks

- Developed a Graph Neural Network (GNN)-based traffic prediction system using Bengaluru traffic data to forecast road congestion and traffic flow.
- Built an interactive Streamlit dashboard with real-time route visualization, live traffic insights, ETA prediction, and alternative route planning using Leaflet and TomTom APIs.
- Implemented data preprocessing, graph construction, and model evaluation using metrics such as MAE, RMSE, and MAPE to measure prediction accuracy.
- Integrated a FastAPI backend for traffic prediction services, enabling dynamic route analysis and scalable deployment for intelligent transportation applications.

Railway Reservation System |

GitHub

Java 21, Spring Boot, Spring Security, PostgreSQL, JPA/Hibernate, Razorpay, Docker

- Developed a full-stack railway reservation system supporting train and station search, schedule filtering, passenger management, fare calculation, seat allocation, booking, and cancellation.
- Implemented JWT authentication, BCrypt password hashing, role-based access control, and integrated a 717-station Indian Railway dataset for station and train management.
- Integrated Razorpay payment processing with booking status management and containerized the application using Docker for deployment.

CERTIFICATIONS

- Google Cloud Engineering – Google Cloud – 2026
- Google UX Design Professional Certificate – Google, Coursera – 2026
- Introduction to Software Engineering – Coursera – 2025
- AWS Academy Graduate – Cloud Foundations – 2025
- Networking Basics – Cisco – 2025